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Masters in Data Science
DSC 540 Data Preparation

TERM PROJECT

MILESTONE 1 - IDENTIFY DATASETS

Due 2026-04-13

PROJECT SUBJECT AREA

The project is a systematic literature *review* of all available randomized controlled trial (RCT) studies pertaining to quantitative outcomes following *primary* anterior cruciate ligament reconstruction (ACLR) surgery in *adult* patients. The purpose was to synthesize evidence from all primary sources of level of evidence I (RCT studies) by way of network meta-analysis to delineate the optimal graft source for ACLR to aid clinicians in their practice.

DATA SOURCES

The data sources were PubMed, Embase, and Web of Science databases for systematic literature review searching. For each database, the search data were pulled via exporting flat CSV files, which was then parsed programmatically via python scripts, XML files from APIs from the databases' institutional access keys, and web scraping using requests module. These methods were used to validate the search results across the methods and across the databases for six individual search strategies, for a total of 54 datasets. Eighteen datasets per each database, and nine datasets per search strategy. The datasets were uploaded onto Harvard Dataverse data repository, using a Creative Commons 4.0 license, and a DOI was granted to *ensure research transparency* and *reproducibility* without sacrificing *integrity*, *credibility* and *authorship*.

Kim, Danny, and Konrad Malinowski. 2026. "What Is the Optimal Graft for Primary Anterior Cruciate Ligament Reconstruction Surgery?" Harvard Dataverse. https://doi.org/10.7910/DVN/NGIW6E.

RELATIONSHIPS

The data from each data source were connected via unique identifier keys, such as DOI number or PMID number. Clinical trial ID numbers were also available for records pulled from clinicaltrials.gov. The columns from each database are shown below. There were a total of 1,309 rows and 30, 42, and 72 columns respectively.

0	id	573	int64
1	Unnamed:	573	int64
2	pmid	573	int64
3	PubDate	573	str
5	Source	573	str
6	AuthorList	573	str
7	LastAuthor	573	str
8	Title	573	str
12	LangList	573	str
13	NlmUniqueID	573	object
16	PubTypeList	573	str
17	RecordStatus	573	str
19	ArticleIds	573	str
21	History	573	str
22	References	573	str
23	HasAbstract	573	int64
24	PmcRefCount	573	int64
25	FullJournalName	573	str
27	SO	573	str
28	source	573	str
11	Pages	572	str
9	Volume	567	object
18	PubStatus	567	str
10	Issue	555	object
14	ISSN	553	str
20	DOI	525	str
15	ESSN	359	str
26	ELocationID	330	str
4	EPubDate	321	str
28	abstract		
29	source		
30	subgroup		

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2	Author Names	228	str
3	Author Addresses	228	str
4	Correspondence Address	145	object
5	Editors	0	float64
6	AiP/IP Entry Date	204	str
7	Full Record Entry Date	303	str
8	Source	317	str
9	Source title	317	str
10	Publication Year	317	int64
11	Volume	230	float64
12	Issue	214	object
13	First Page	171	object
14	Last Page	158	object
15	Date of Publication	317	str
16	Publication Type	317	str
17	Conference Name	30	object
18	Conference Location	30	object
19	Conference Date	30	object
20	Conference Editors	0	float64
21	ISSN	237	str
22	ISBN	0	float64
23	Book Publisher	254	str
24	Abstract	315	str
25	Original Abstract	0	float64
26	Author Keywords	229	str
27	Emtree Drug Index Terms (Major Focus)	35	object
28	Emtree Drug Index Terms	65	object
29	Emtree Medical Index Terms (Major Focus)	312	str
30	Emtree Medical Index Terms	317	str
31	Drug Tradenames	2	object
32	Drug Manufacturer	1	object
33	Device Tradenames	41	str
34	Device Manufacturer	38	str
35	CAS Registry Numbers	38	object
36	Molecular Sequence Numbers	0	float64
37	Embase Classification	97	str
38	Clinical Trial Numbers	49	object
39	Article Language	317	str
40	Summary Language	291	str
41	Embase Accession ID	40	float64
42	Medline PMID	171	float64
43	PUI	317	str
44	DOI	207	str
45	Full Text Link	285	str
46	Embase Link	317	str
47	Open URL Link	239	str

48	Copyright	317	str
49	source	317	str

0	PT	516	str	Publication Type
1	AU	516	str	Authors
2	BA	0	float64	Book Authors
3	BE	0	float64	Book Editors
4	GP	0	float64	Book Group Authors
5	AF	516	str	Author Full Names
6	BF	0	float64	Book Author Full Names
7	CA	8	object	Group Authors
8	TI	516	str	Article Title
9	SO	516	str	Source Title
10	SE	0	float64	Book Series Title
11	BS	0	float64	Book Series Subtitle
12	LA	516	str	Language
13	DT	516	str	Document Type
14	CT	5	object	Conference Title
15	CY	5	object	Conference Date
16	CL	5	object	Conference Location
17	SP	3	object	Conference Sponsor
18	HO	0	float64	Conference Host
19	DE	413	str	Author Keywords
20	ID	494	str	Keywords Plus
21	AB	512	str	Abstract
22	C1	511	str	Addresses
23	C3	469	str	Affiliations
24	RP	512	str	Reprint Addresses
25	EM	494	str	Email Addresses
26	RI	350	str	Researcher Ids
27	OI	366	str	ORCIDs
28	FU	188	str	Funding Orgs
29	FP	185	str	Funding Name Preferred
30	FX	188	str	Funding Text
31	CR	323	object	Cited References
32	NR	516	int64	Cited Reference Count
33	TC	516	int64	Times Cited, WoS Core
34	Z9	516	int64	Times Cited, All Databases
35	U1	516	int64	180 Day Usage Count
36	U2	516	int64	Since 2013 Usage Count
37	PU	516	str	Publisher
38	PI	516	str	Publisher City
39	PA	516	str	Publisher Address
40	SN	437	str	ISSN
41	EI	418	str	eISSN
42	BN	0	float64	ISBN
43	J9	516	str	Journal Abbreviation
44	JI	516	str	Journal ISO Abbreviation
45	PD	500	str	Publication Date

46	PY	516	int64	Publication Year
47	VL	499	object	Volume
48	IS	477	float64	Issue
49	PN	0	float64	Part Number
50	SU	2	object	Supplement
51	SI	0	float64	Special Issue
52	MA	0	float64	Meeting Abstract
53	BP	355	object	Start Page
54	EP	355	object	End Page
55	AR	163	object	Article Number
56	DI	507	str	DOI
57	DL	507	str	DOI Link
58	D2	0	float64	Book DOI
59	EA	107	str	Early Access Date
60	PG	516	int64	Number of Pages
61	WC	516	str	WoS Categories
62	WE	516	str	Web of Science Index
63	SC	516	str	Research Areas
64	GA	516	str	IDS Number
65	PM	498	float64	Pubmed Id
66	OA	246	str	Open Access Designations
67	HC	0	float64	Highly Cited Status
68	HP	0	float64	Hot Paper Status
69	DA	516	str	Date of Export
70	UT	516	str	UT (Unique WOS ID)
71	source	516	str	Web of Science Record

PROJECT APPROACH/PLAN

Word count: 506

The search strategy was designed according to the *PICOS* format:

Table 1 PICOs format for identifying datasets from database searching

P	Population	Skeletally-mature adult patients[1]
I	Intervention	Primary anterior cruciate ligament reconstruction surgery
C	Comparator	Four autologous and two allogenic grafts
O	Outcomes	Patient-reported outcome measures, objective measures of knee stability, and adverse events and/or complications
S	Study design	Randomized controlled trials

Literature searches using the search strategies (i.e., queries) for each of the three databases were performed for each of the six graft types most commonly harvested from the tendon of the patients' *ipsilateral* limb: bone-patellar tendon-bone, hamstring (single, double, or quadruple-stranded semitendinosus ± gracilis), quadriceps (all soft-tissue or with patellar bone i.e., "with bony attachment"), peroneus longus (with or without peroneus brevis tenodesis), and the two allogenic grafts, achilles and tibialis (anterior and posterior).

[1] defined as mean age ≥ 18.0 years of all included patients in each particular study

The systematic approach employed was to design the query to encapsulate the eligibility criteria domains of population, interventions, outcomes and study design and then comparators by subsequently apply this "template" for each of the six subgroups for a total of six datasets (CSV files) for NCBI's PubMed. Then the search strategy was "translated" into the appropriate query syntax particular to Elsevier's Embase[2] and Clarivate's Web of Science Core Collection. This was achieved by writing a python script using regular expressions (regex) to find and replace the appropriate characters and field tag specifications that differentiate the query syntax across databases. A total of 18 datasets will be downloaded for each data source type: 18 CSV files downloaded directly from the three databases' export functions, 18 CSV files parsed from XML using the databases' API using python's `requests` module, and 18 CSV files parsed from the HTML file scraped using python's `beautifulsoup4` module, for a total of 48 dataset files.

CONCERNS/CHALLENGES

The concerns and challenges were that there were massive discrepancies between the datasets by the methods in which they were pulled. Despite search strategies remaining constant across the dates of searches performed, the datasets remained inconsistent and mutable in both field structure, number of records, and the values themselves. In order to resolve these discrepancies, the methodology and systems engineering approach itself became the focal point of improvement, not the datasets themselves with the aim of implementing the refined python scripts to pull data at a later stage in time, as is required regardless prior to submission of the manuscript. In case of any changes in the analysis results themselves, a plan was made to resolve such differences at the time of the update search and data collection routinely done at the late stage.

ETHICAL IMPLICATION

The ethical implications are that if the update data collection remains inconsistent with true values, then the output of the analyses and conclusions extrapolated from them, will be cause for retraction of the submitted article. As the protocol was registered and published publicly and delineates the process by which this project was carried out, transparency, integrity and reproducibility remained aligned with ideals and values of the team and respective ethical committees that oversee these types of study designs.